

Abhishek Maurya

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PROFILE

Research-driven AI/ML Developer focused on architecting production-grade Generative AI pipelines and robust microservices. Strong proponent of comprehensive system visualization and theoretical planning prior to execution, enabling the autonomous delivery of end-to-end, crash-resilient solutions. Driven by complex technical challenges with a clear vision to build scalable, high-impact core technologies.

EDUCATION

B.Sc. in Information Technology – Tolani College of Commerce, Mumbai University (2022–2025) CGPA: 8.0/10 |

Relevant Coursework: *Linear Algebra, Statistics, Probability*

SKILLS

- **Generative AI & LLMs:** LangChain, LangGraph, Hugging Face, Gemini, FAISS, Prompt Engineering, Fine-tuning (PEFT, QLoRA, Unsloth), RAG Pipelines
- **Machine Learning:** Scikit-learn, Regression, Classification, Clustering, Feature Engineering
- **Deep Learning:** TensorFlow, PyTorch, CNN, ANN
- **Backend & MLOps:** FastAPI, Flask, Docker, MLflow, DVC, CI/CD, REST APIs, Async Processing.
- **Programming & Data:** Python, SQL, MongoDB, NumPy, Pandas, Matplotlib
- **Tools & Visualization:** Git, GitHub, Streamlit, Manim CE, Claude Code.

WORK EXPERIENCE

AI Intern – Nullclass (Remote) | June 2025 – August 2025

- Designed and delivered **7 self-built, production-grade AI chatbot projects** in just 1 month during a 3-month internship.
- Built systems using **RAG pipelines, multilingual NLP, vector search**, and **LLM-based reasoning** using Hugging Face & LangChain.
- Implemented a **medical Q&A bot** using the MedQuAD dataset with translation support and chunk-based retrieval.
- Developed **dynamic knowledge base expansion** via automated web scraping and vector DB updates.
- Extended capabilities with **sentiment-aware response systems** and **multimodal chat using Gemini** (text + image input/output).
- Employed tools like **FastAPI, FAISS, Deep Translator**, and **Google Gemini**, focusing on scalable, real-world deployments.

PROJECTS

1. AutoShorts: Automated AI Video Generation Pipeline | Python, LangGraph, Gemini, Manim CE, ElevenLabs. | [GitHub repo](#).

- **Pipeline Architecture:** Architected a **4-phase asynchronous microservice pipeline** automating the entire video creation lifecycle, from **autonomous topic discovery** and web scraping to dynamic script generation and **3D rendering**.
- **Audio-Visual Sync (Under-run Gate):** Engineered an '**Under-run Gate**' utilizing Pydub for **in-memory silence trimming** (dBFS thresholding), calculating precise millisecond timelines to guarantee **flawless frame-perfect audio-visual synchronizatio**
- **State Management & Resilience:** Implemented strict state management via **atomic file operations** (os.replace) and job-centric isolation, ensuring a **100% crash-resilient architecture** capable of seamless resumption from any point of failure.
- **Efficiency & Cost Optimization:** Optimized operational costs by developing a **SHA-256 hash-based caching mechanism** (idempotency) to eliminate redundant LLM calls, alongside **asyncio.Semaphore to manage third-party API rate limits** securely.search

2. Vehicle Insurance MLOps Pipeline | Python, Scikit-learn, DVC, Git, MongoDB

- **Infrastructure & Automation:** Developed an **end-to-end MLOps pipeline** for vehicle insurance claim prediction, utilizing modular Python scripts to ensure **100% reproducibility** across development environments.
- **Data & Model Versioning:** Integrated **Git and Data Version Control (DVC)** for rigorous tracking of large-scale datasets and model artifacts, **enabling seamless** and structured experimentation cycles.
- **System Scalability:** Architected a scalable infrastructure utilizing **MongoDB** for efficient data handling, **decoupling the data** ingestion and model training phases to **maximize system throughput**.

CERTIFICATIONS & INTERNSHIPS

- [Internship Certificate](#) – Novadita Infotech (2025) -Awarded for completing 3 basic ML projects under mentorship.
- [Internship Certificate](#) – Nullclass 3-month AI internship building 7 production-grade GenAI projects using LLMs, LangChain, and FastAPI.
- [Course Completion](#) – Nullclass (2025) - Successfully completed an AI/ML course covering LLMs, NLP, and RAG workflows.